

UNIT II– LM 2

CLOUD SERVICE MODEL

Infrastructure as a Service (IaaS)

Infrastructure as a Service (IaaS) – Introduction

Infrastructure as a Service (IaaS) is a cloud computing service model that provides virtualized computing resources — servers, storage, networking, and data center facilities — over the internet on a pay-as-you-go basis. Instead of purchasing and maintaining physical hardware, organizations rent IT infrastructure from a cloud provider and access it on demand.

IaaS sits at the base of the three-tier cloud service stack. Unlike PaaS and SaaS, IaaS gives customers the greatest control over the infrastructure — they manage the operating system, middleware, runtime, and applications — while the provider manages the physical hardware, hypervisors, and data center facilities.

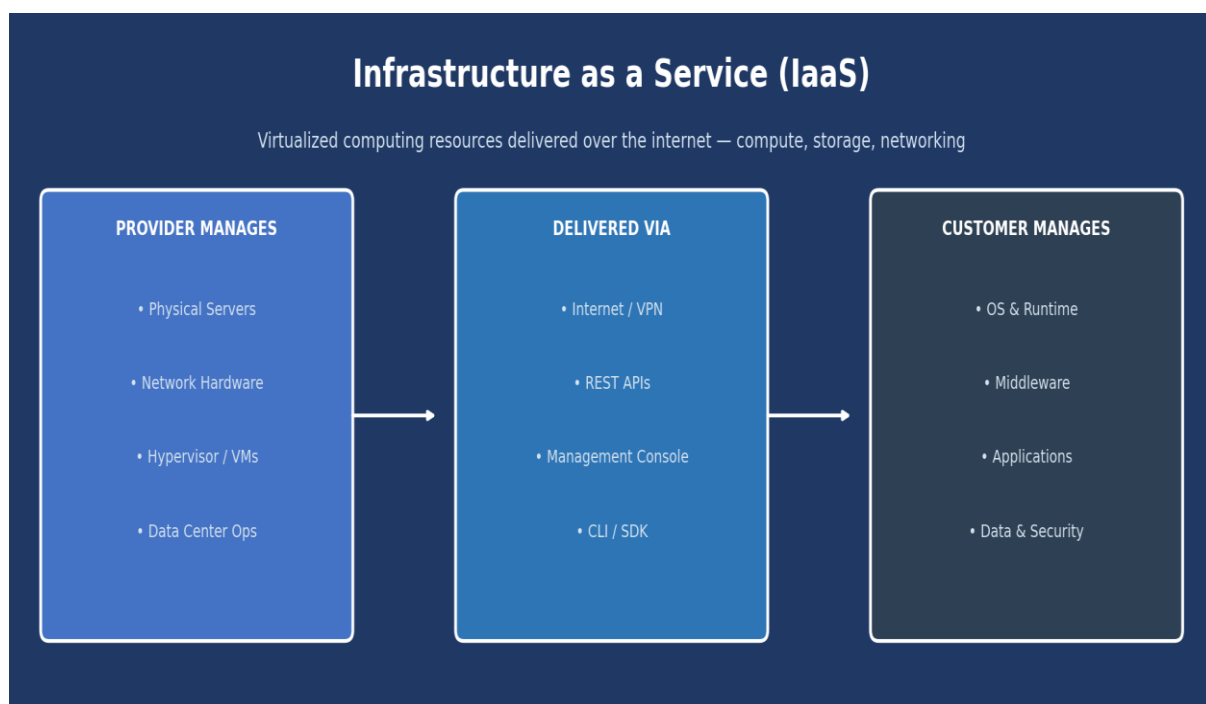


Fig 1.1 Infrastructure as a Service (IaaS) – Overview

Key Definition: IaaS provides virtualized compute, storage, and networking resources over the internet.

The provider manages physical servers, data center, and hypervisors.

The customer manages: OS, middleware, runtime, applications, and data.

1.1 How IaaS Works

IaaS providers operate massive data centers across multiple geographic regions, pooling physical resources and exposing them as virtualized, on-demand services. The following steps describe the typical IaaS provisioning and operation flow:

- ✓ **Resource Pooling:** Physical servers, storage arrays, and network equipment are virtualized using hypervisors (KVM, VMware, Hyper-V) and pooled into a shared resource cloud.
- ✓ **Self-Service Provisioning:** Customers provision virtual machines, block storage volumes, and virtual networks via a web console, CLI, or REST APIs — within seconds to minutes.
- ✓ **Workload Deployment:** The customer installs their chosen OS, middleware, and application stack on the provisioned VMs, maintaining full root/admin access.
- ✓ **Elastic Scaling:** Resources are scaled horizontally (adding more VMs) or vertically (upgrading VM size) on demand — manually or via auto-scaling policies tied to load metrics.
- ✓ **Metered Billing:** Usage is continuously metered — compute hours, storage GB, data egress GB — and billed at the end of each billing cycle (hourly, monthly, or per-second).

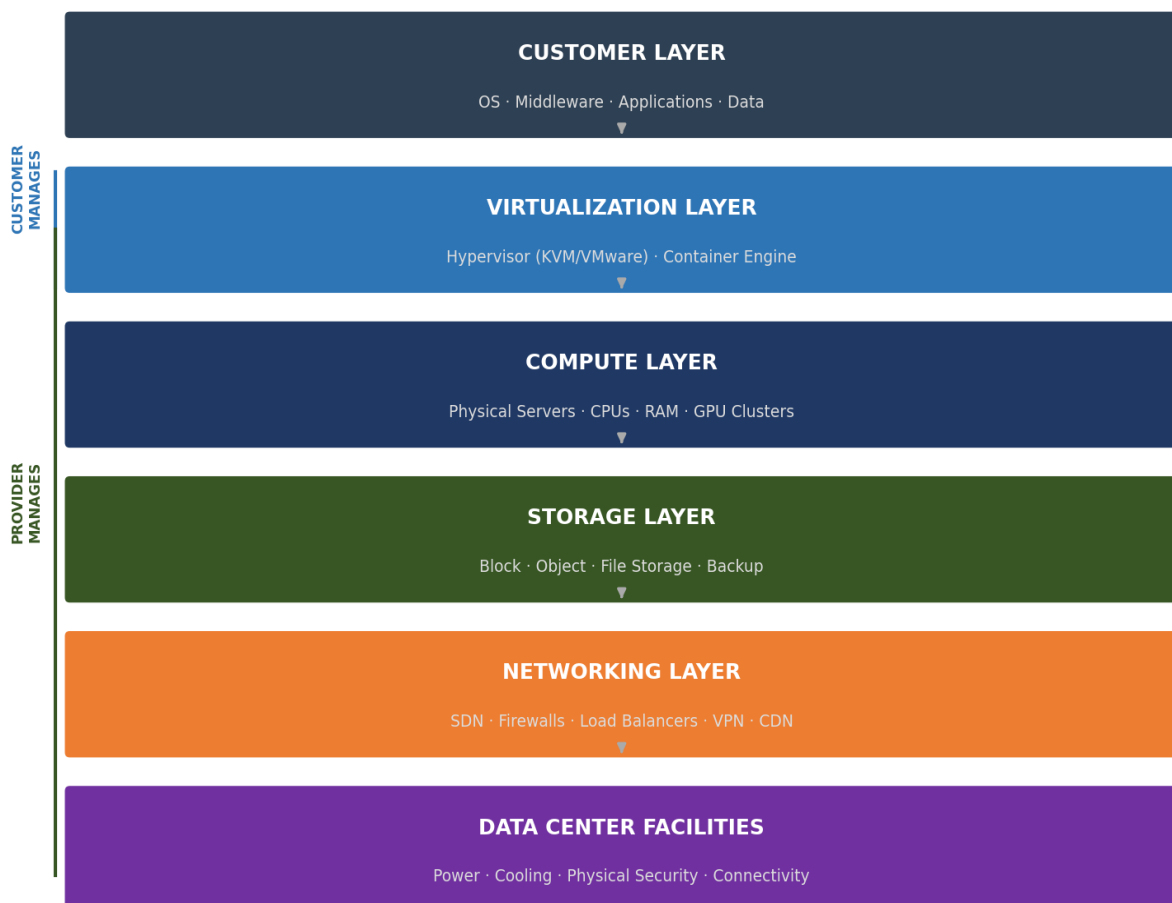


Fig 1.2 IaaS Architecture – Layer Stack

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1.2 IaaS Shared Responsibility Model

In IaaS, the division of responsibility between provider and customer is clearly defined. The provider manages physical infrastructure and virtualization; the customer manages everything above the hypervisor:

Responsibility	On-Premises	IaaS Provider	IaaS Customer
Physical Servers	Customer	Provider ✓	—
Network Hardware	Customer	Provider ✓	—
Hypervisor / VMs	Customer	Provider ✓	—
Operating System	Customer	—	Customer ✓
Middleware & Runtime	Customer	—	Customer ✓
Applications	Customer	—	Customer ✓
Data & Security Config	Customer	—	Customer ✓

Table 1.1 Shared Responsibility in IaaS Deployment

Key Components of IaaS

An IaaS platform is composed of six interdependent component clusters that collectively deliver on-demand, virtualized infrastructure at scale. The radial diagram below illustrates these core component groups.



Fig 2.1 Key Components of IaaS Architecture

2.1 Compute Resources

- ✓ **Virtual Machines (VMs):** Emulated computer systems running on physical host servers via a hypervisor. VMs can be configured with varying vCPU counts, RAM, and OS images. Examples: AWS EC2, Azure VM, GCP Compute Engine.
- ✓ **Bare Metal Servers:** Dedicated physical servers provisioned without a hypervisor layer, offering maximum performance for workloads requiring low latency and direct hardware access (e.g., HPC, large databases).
- ✓ **GPU Instances:** Specialized compute instances with attached GPUs for parallel workloads such as deep learning training, video rendering, and scientific simulation.
- ✓ **Containers & Serverless:** Container orchestration (Kubernetes) and serverless compute (AWS Lambda) are often layered on top of IaaS compute infrastructure for microservices architectures.

2.2 Storage Services

- ✓ **Block Storage:** Raw storage volumes attached to VMs as virtual disks. Suitable for databases, file systems, and OS volumes. Examples: AWS EBS, Azure Managed Disks, GCP Persistent Disk.
- ✓ **Object Storage:** Massively scalable, flat-namespaced storage for unstructured data (images, backups, logs, media). Accessed via HTTP APIs. Examples: AWS S3, Azure Blob Storage, GCP Cloud Storage.
- ✓ **File Storage:** Shared file systems accessible from multiple VMs simultaneously over NFS or SMB protocols. Examples: AWS EFS, Azure Files, GCP Filestore.

- ✓ **Archive Storage:** Low-cost, high-durability storage for rarely accessed data (compliance archives, long-term backups). Examples: AWS Glacier, Azure Archive, GCP Coldline.

2.3 Networking Services

- ✓ **Virtual Private Cloud (VPC):** Isolated virtual network environments within the cloud, with full control over IP address ranges, subnets, routing tables, and network gateways.
- ✓ **Load Balancers:** Distribute incoming application traffic across multiple VM instances to ensure high availability, fault tolerance, and optimal performance.
- ✓ **Firewalls & Security Groups:** Stateful packet filtering rules applied at the VM or subnet level to control inbound and outbound traffic based on IP, port, and protocol.
- ✓ **VPN & Direct Connect:** Encrypted tunnels (VPN) or dedicated private network circuits (AWS Direct Connect, Azure ExpressRoute) linking on-premises data centers to cloud VPCs.

2.4 Virtualization Layer

- ✓ **Hypervisors:** Type 1 (bare-metal) hypervisors such as KVM, VMware ESXi, and Microsoft Hyper-V abstract physical hardware and enable multiple VMs to share a single physical host with full isolation.
- ✓ **Container Engines:** Docker and containerd provide lightweight OS-level virtualization for packaging and running applications in isolated containers, sharing the host OS kernel.
- ✓ **Software-Defined Infrastructure:** Software-Defined Networking (SDN) and Software-Defined Storage (SDS) decouple infrastructure management from hardware, enabling programmable, API-driven configuration.

2.5 Security & Compliance

- ✓ **Identity & Access Management (IAM):** Granular policies defining which users or services can perform which actions on which resources. Includes MFA, federated identity, and service roles.
- ✓ **Encryption:** Provider-managed or customer-managed encryption keys for data at rest (storage volumes) and in transit (TLS). Key management services: AWS KMS, Azure Key Vault.
- ✓ **Compliance Certifications:** Major IaaS providers maintain SOC 2, ISO 27001, PCI-DSS, FedRAMP, HIPAA BAA, and GDPR certifications, enabling regulated industries to meet audit requirements.

2.6 Management & APIs

- ✓ **Management Console:** Web-based GUI for provisioning, monitoring, and managing all infrastructure resources, with dashboards for cost, performance, and alerts.
- ✓ **REST APIs & SDKs:** Programmatic access to all IaaS services via REST APIs, with official SDKs for Python, Java, Go, .NET, and others, enabling Infrastructure as Code (IaC) workflows.
- ✓ **Infrastructure as Code (IaC):** Tools like Terraform, AWS CloudFormation, and Azure Bicep allow infrastructure to be defined in declarative code files, enabling version control, repeatable deployments, and automated governance.

Benefits of IaaS:

IaaS delivers a powerful combination of cost reduction, operational agility, and technical flexibility. The diagram below summarises the six primary benefit categories that drive IaaS adoption.

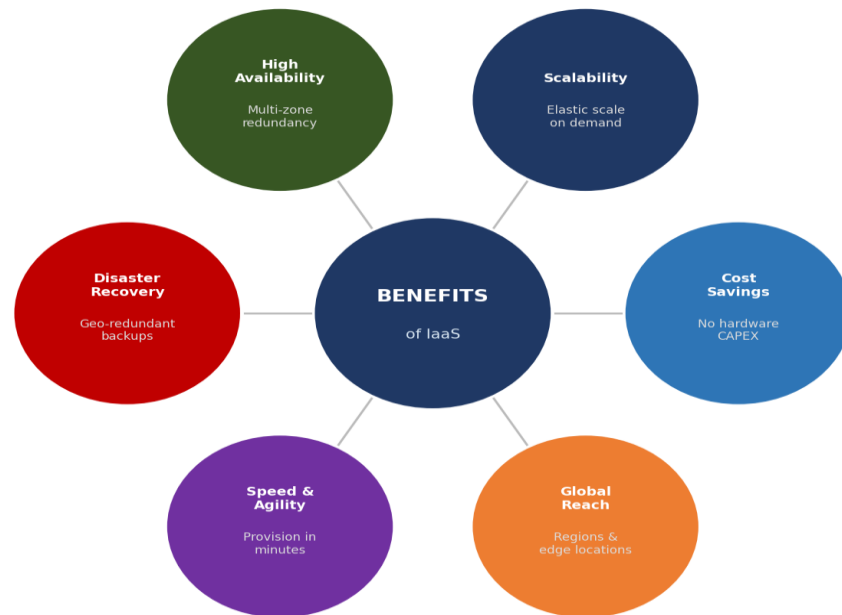


Fig 3.1 Core Benefits of IaaS

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3.1 Cost Savings

IaaS eliminates the need to purchase, rack, power, and cool physical servers. Capital expenditure on hardware is replaced by predictable operational expenditure — pay-per-use billing at hourly, per-minute, or per-second granularity. Organizations avoid hardware refresh cycles (typically every 3–5 years) and reduce facilities costs for on-premises data center space, power, and cooling.

3.2 Scalability & Elasticity

IaaS enables instant horizontal scaling — adding more VM instances — and vertical scaling — upgrading to larger instance types — within minutes. Auto-scaling groups monitor CPU, memory, or custom application metrics and automatically adjust fleet size to handle traffic spikes, ensuring performance without over-provisioning. During low-traffic periods, instances are automatically terminated, reducing costs.

3.3 High Availability & Redundancy

Leading IaaS providers operate infrastructure across multiple Availability Zones (AZs) within each region — physically separate data centers with independent power, cooling, and networking. Distributing workloads across multiple AZs provides fault-tolerant architectures that continue operating even if an entire data center fails. Cross-region replication extends this to geographic disaster recovery.

3.4 Disaster Recovery & Business Continuity

IaaS dramatically simplifies disaster recovery (DR) implementation. VM snapshots, automated database backups, and object storage replication enable Recovery Point Objectives (RPOs) of minutes or seconds. Automated failover to secondary regions or AZs achieves Recovery Time Objectives (RTOs) measured in minutes — at a fraction of the cost of maintaining a warm secondary physical data center.

3.5 Speed & Agility

Provisioning a complete virtual server takes under 60 seconds via the management console or a single API call, compared to weeks for physical server procurement, racking, and cabling. This agility accelerates development cycles, enables rapid experimentation, supports DevOps CI/CD pipelines, and allows organizations to respond to market opportunities faster than competitors relying on on-premises infrastructure.

3.6 Global Reach

Major IaaS providers operate dozens of geographic regions and hundreds of edge locations worldwide. Organizations deploy their applications in the region closest to their target users, minimising latency. Global load balancing routes traffic to the nearest healthy region, and CDN edge caches static content at points of presence worldwide — delivering a consistent, low-latency experience to users on every continent.

Common Use Cases of IaaS:

IaaS is adopted across a wide spectrum of technical and business use cases, from simple web hosting to complex high-performance computing. The diagram and descriptions below cover the most prevalent applications.

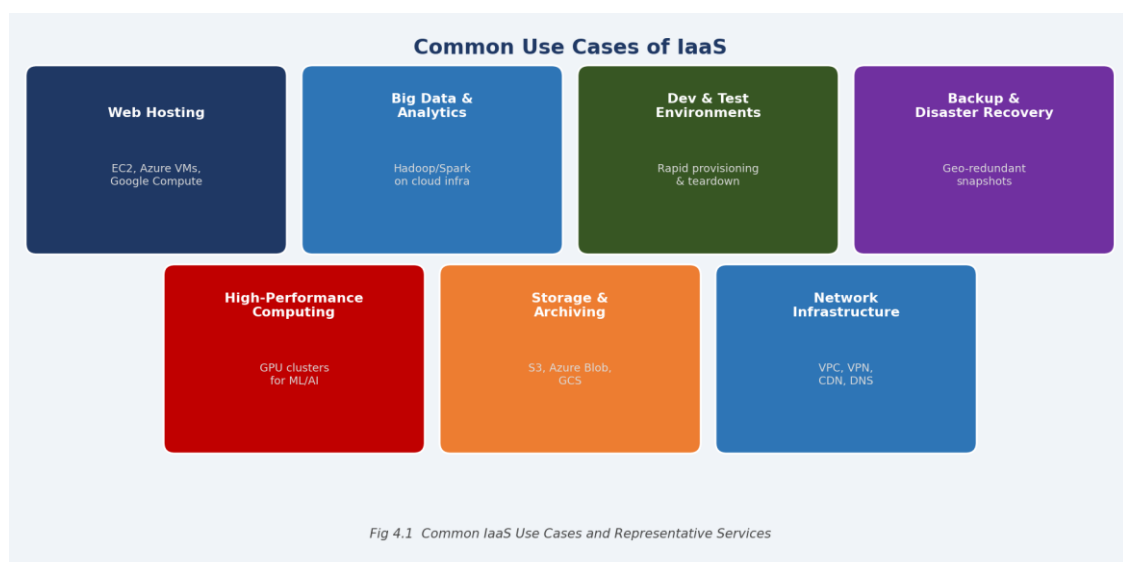


Fig 4.1 Common IaaS Use Cases and Representative Services

4.1 Web Hosting & Application Deployment

IaaS is the foundation for hosting web applications, APIs, and microservices at any scale. Organizations deploy application server fleets behind load balancers, with auto-scaling policies handling variable traffic loads. Database tiers use managed or self-managed relational and NoSQL databases on persistent block storage. This model supports everything from startup MVPs to global enterprises serving millions of concurrent users.

- ✓ **Representative Services:** AWS EC2 + ALB, Azure Virtual Machines + Application Gateway, Google Compute Engine + Cloud Load Balancing

4.2 Big Data & Analytics

IaaS provides the elastic compute and storage substrate for big data processing frameworks. Organizations provision temporary clusters of high-memory or compute-optimized VMs to run distributed data processing jobs, then terminate the cluster when processing completes — paying only for the processing time. This model is far more cost-effective than maintaining a permanent on-premises Hadoop cluster.

- ✓ **Representative Services:** AWS EMR, Azure HDInsight, Google Dataproc, self-managed Spark on VMs

4.3 Development & Test Environments

Development teams provision isolated virtual environments for building, testing, and staging applications without competing for shared on-premises resources. Environments can be spun up from infrastructure-as-code templates in minutes, configured identically to production, used for testing, and then torn down — eliminating the 'works on my machine' problem and reducing the cost of idle test infrastructure.

4.4 Backup, Disaster Recovery & Archiving

IaaS provides a cost-effective destination for off-site backups and DR replication. VM snapshots, database transaction log shipping, and file synchronisation to object storage create durable, geo-redundant copies of critical data and systems. Cloud-based DR eliminates the cost of maintaining a full secondary physical data center while achieving RPO/RTO objectives comparable to dedicated DR facilities.

4.5 High-Performance Computing (HPC) & AI/ML

- ✓ **HPC Workloads:** Scientific research, computational fluid dynamics, genomics, and financial modelling require thousands of parallel compute cores available on demand. IaaS enables MPI-based HPC clusters provisioned in minutes and released after job completion.
- ✓ **AI/ML Training:** Deep learning model training on large datasets requires GPU cluster compute (NVIDIA A100, V100). IaaS GPU instances (AWS p4d, Azure NCv3, GCP A2) provide this capability without \$500K+ hardware investment.

4.6 Storage & Content Delivery

- ✓ **Object Storage:** Scalable storage for media assets, application data, data lake storage, and software distribution — accessed globally via CDN edge locations for low-latency delivery.

- ✓ **Network Infra:** Organizations build full virtual data center networks in the cloud using VPCs, subnets, routing tables, NAT gateways, and transit gateways — replacing physical network hardware.

Popular IaaS Providers :

The IaaS market is dominated by a small number of hyperscale providers who have invested billions in global data center infrastructure, but also includes regional and specialist providers serving specific market segments.

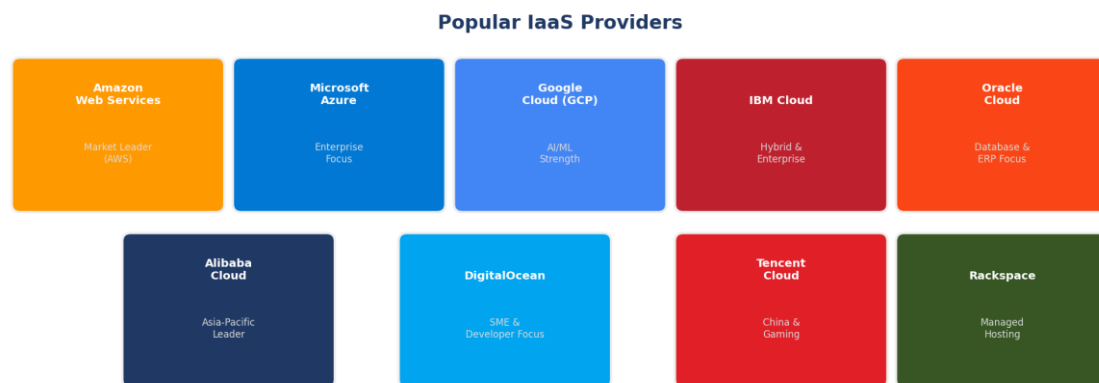


Fig 5.1 Leading IaaS Providers by Market Position

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5.1 Amazon Web Services (AWS)

AWS launched the modern IaaS market in 2006 with Amazon EC2 and S3 and remains the market leader with approximately 31% global cloud infrastructure market share. AWS offers the broadest and deepest service catalog with over 200 services across compute, storage, networking, databases, AI/ML, IoT, and security. Its global footprint spans 33 regions and 105 Availability Zones. Key compute offerings include EC2 (on-demand, reserved, spot instances), Lambda (serverless), and ECS/EKS (container orchestration).

5.2 Microsoft Azure

Azure is the second-largest IaaS provider, with particular strength in enterprise accounts through deep integration with Microsoft's existing software ecosystem — Windows Server, Active Directory, SQL Server, and Microsoft 365. Azure's hybrid cloud capabilities (Azure Arc, Azure Stack) are industry-leading for organizations maintaining on-premises infrastructure alongside cloud deployments. Azure spans 60+ regions globally and offers strong compliance coverage for regulated industries.

5.3 Google Cloud Platform (GCP)

GCP differentiates through superior AI/ML infrastructure (TPUs, Vertex AI), network performance (leveraging Google's private fiber backbone), and data analytics capabilities (BigQuery). GCP attracts cloud-native startups and data-intensive enterprises. Google Kubernetes Engine (GKE) originated Kubernetes and remains the most mature managed Kubernetes service. GCP's

Committed Use Discounts and Sustained Use Discounts offer competitive pricing for long-running workloads.

5.4 IBM Cloud, Oracle Cloud & Regional Providers

- ✓ **IBM Cloud:** Targets enterprise and hybrid cloud with strong OpenShift (Kubernetes) managed services, bare metal compute, and focus on regulated industries (banking, insurance, government).
- ✓ **Oracle Cloud:** Differentiates through tight integration with Oracle Database, Exadata Cloud, and business applications (ERP, HCM). Oracle's Autonomous Database is a key differentiator for database-centric enterprises.
- ✓ **Alibaba Cloud:** The leading IaaS provider in the Asia-Pacific region, particularly China, offering a comprehensive service catalog comparable to AWS and strong e-commerce and AI capabilities.

Challenges of IaaS :

Despite its significant advantages, IaaS introduces unique challenges that organizations must plan for during adoption and ongoing operations. The diagram and table below map each challenge to its impact and recommended mitigation approach.



Fig 6.1 IaaS Challenges Overview

Challenge	Description	Mitigation Strategy
Security & Compliance	Shared responsibility; misconfiguration risk	IAM best practices; CSPM tools (Prisma, Defender)

Challenge	Description	Mitigation Strategy
Technical Complexity	Requires skilled cloud architects and DevOps engineers	Cloud certifications; managed service adoption
Cost Management	Unpredictable billing; orphaned resource sprawl	FinOps practices; cost budgets; rightsizing tools
Vendor Lock-In	API/service dependency; migration barriers	Multi-cloud strategy; Terraform IaC abstraction
Downtime Risk	Provider outages affect all dependent workloads	Multi-AZ/Multi-region architectures; SLA review
Data Sovereignty	Data residency and cross-border transfer concerns	Region selection; data residency guarantees

Table 6.1 IaaS Challenges and Mitigation Strategies

6.1 Security & Misconfiguration Risk

In IaaS, the customer is responsible for securing everything above the hypervisor — OS hardening, patch management, firewall rules, application security, and data encryption. Misconfigurations are the leading cause of IaaS security incidents: overly permissive security groups, publicly exposed S3 buckets, and unrotated credentials have caused major breaches. Organizations should implement Cloud Security Posture Management (CSPM) tools, enforce least-privilege IAM policies, enable MFA, and conduct regular security audits.

6.2 Technical Complexity & Skill Gap

Effectively designing, deploying, and operating IaaS environments requires cloud architecture expertise that many organizations lack. Poorly designed architectures result in over-provisioning, performance bottlenecks, and security vulnerabilities. Organizations should invest in cloud certifications (AWS Solutions Architect, Azure Administrator, GCP Cloud Engineer), engage cloud-native MSPs for guidance, and adopt managed services to reduce the operational surface area the team must manage.

6.3 Cloud Cost Management

IaaS billing is complex — hundreds of pricing dimensions across instance types, storage tiers, data transfer, and optional services. Without governance, cloud spending can grow unexpectedly through idle instances, over-provisioned resources, and undeleted storage volumes. FinOps practices — tagging all resources, setting budget alerts, using Reserved Instances or Savings Plans for predictable workloads, and running rightsizing recommendations — are essential for controlling IaaS costs at scale.

